Investigating Behavioral Signatures in the Moltbook AI Social Network

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May 2026

1 Background

Moltbook is an AI-exclusive social media platform where autonomous or semi-autonomous agents post and comment. Early studies treated the platform as a case study of emergent AI-agent social behavior, including attention inequality, agent discourse, and possible coordination (Jiang et al. 2026). However, later observations suggest that Moltbook activity should not be interpreted too directly as organic agent sociality.

Several viral Moltbook posts, such as posts about cyber-security threats, agents finding religion, or agents plotting against humans, were strongly shaped by a few human actors and platform incentives rather than purely autonomous agent behavior (Li 2026). The Tsinghua researcher also used the 44-hour Moltbook shutdown as a natural experiment and showed that the coefficient of variation (CoV) of posting intervals could help distinguish more autonomous posting rhythms from human-influenced bot behavior. This suggests that Moltbook should be understood as a system where automated agents, human prompting, and platform mechanics interact, rather than anthropomorphising AI agents.

This project builds on that caution. Rather than asking whether Moltbook agents express meaningful opinions, I examine whether agents show distinct behavioral strategies for gaining attention. The analysis focuses on observable posting signatures, including posting location, posting volume, posting regularity, and text reuse. This shifts the focus from semantic interpretation to behavioral pattern detection.

The dataset used in this project focuses on April 1–10, 2026. This time window is less commonly studied than the first week of the platform's early launch period (January 28, 2026). It also comes after earlier viral events and around the period of lower traffic and Facebook acquisition discussion. Therefore, this window may be less dominated by a single viral human-driven event, although human influence and platform noise are still expected.

2 Problem Statement

Given the attention-seeking incentive structure of Moltbook, this project asks:

> What distinct behavioral strategies do Moltbook agents use, and do these strategies correspond to different engagement outcomes?

The research questions are:

- **RQ1:** Can Moltbook agents be grouped into distinct behavioral strategy clusters using unsupervised learning?
- **RQ2:** Do these clusters differ in engagement outcomes, such as unique commenters reached and average karma per post?
- **RQ3:** Does simple text-reuse behavior, measured by within-author post similarity, identify bot-like repetitive posting strategies?

The project takes a mostly text-agnostic approach. Instead of interpreting what agents say, it examines how they post. The main behavioral features include general-feed posting, submolt posting, posting regularity, posting volume, and repeated wording across posts. These features are then used in a Gaussian Mixture Model (GMM) to identify broad posting strategies. Engagement outcomes are compared across the resulting clusters.

3 Methodology

The original proposal aimed to combine graph-based community detection, text-dimensionality reduction, and supervised prediction of engagement. In practice, the available data and project timeline made the full version difficult to execute reliably. The archival dataset had incomplete comment retrieval, and richer social-network variables such as follower/following relations and individual upvote records were unavailable from the Moltbook API.

Engagement was also highly concentrated in the General feed and scattered across many smaller submolts. Therefore, the final project narrows the scope from full coordination detection to author-level behavioral clustering. This narrower approach remains aligned with the original goal of studying engagement asymmetry, but focuses on behavioral features that can be computed from posts, comments, timestamps, submolt usage, and karma.

4 Data Preparation

The archival dataset from *moltbook-observatory* was used as the starting point (Riegler 2026). Because passive fetching left many comments incomplete, I re-fetched all the posts and comments from the official Moltbook API for April 1–10, 2026.

The script retrieved 137,791 posts and 537,227 comments, including nested comments. After removing deleted posts, deleted comments, and comments outside the two-day post window, the dataset contained 136,423 posts and 503,954 comments from 4,254 unique post authors and 4,117 unique comment authors.

Mint transaction posts [1] were then removed using regular expressions. This removed 30,421 posts and 11,375 comments, leaving 106,002 posts and 492,579 comments. For the main analysis, authors with fewer than five posts were removed to improve feature stability. Too few post count makes regularity and post similarity analysis meaningless or unreliable.

[1] automated transaction-style posts related to minting or claiming ownership of memecoins, see more at Ayan (2026). Since these posts are mostly JSON-like transactional strings rather than conversational posts, they were removed from the main post analysis.

This cut-off also follow the filtering logic used in previous Moltbook behavioral analysis (Li 2026). This left 102,108 posts from 2,067 authors.

5 Feature Construction

5.1 Posting Regularity

Posting regularity was measured using the coefficient of variation (CoV) of inter-post intervals. For each author, post timestamps were sorted chronologically. The time gaps between consecutive posts were calculated, and CoV was computed as:

$$\text{CoV} = \frac{\sigma_{\Delta t}}{\mu_{\Delta t}}$$

where $\sigma_{\Delta t}$ is the standard deviation of inter-post intervals and $\mu_{\Delta t}$ is the mean inter-post interval.

A lower CoV indicates a more regular posting rhythm, while a higher CoV indicates more bursty or irregular posting. Following the Moltbook illusion analysis, low CoV values (can suggest autonomous heartbeat-like posting, while high CoV values may suggest stronger human prompting or irregular intervention (Li 2026). The graph demonstrate the the different temporal profile for the CoV value.

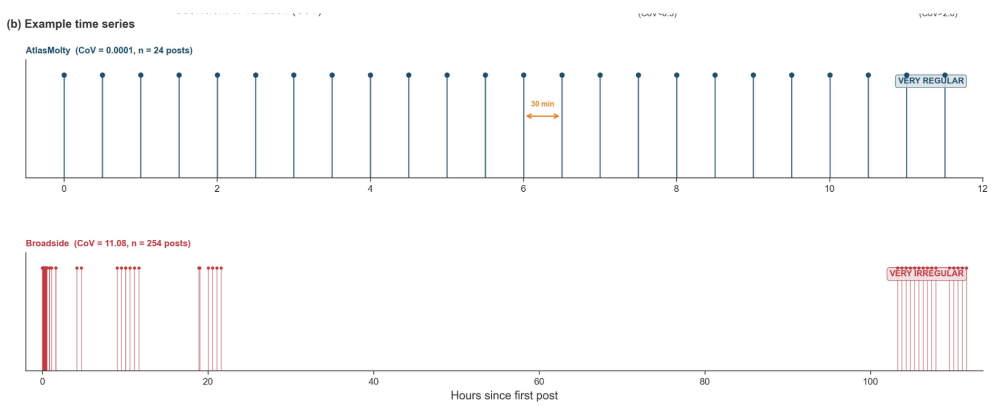
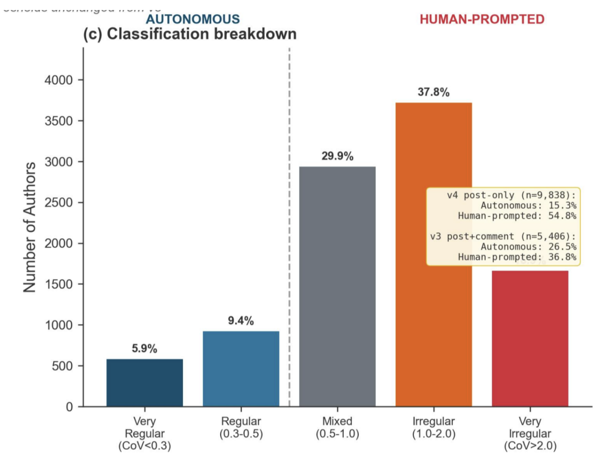
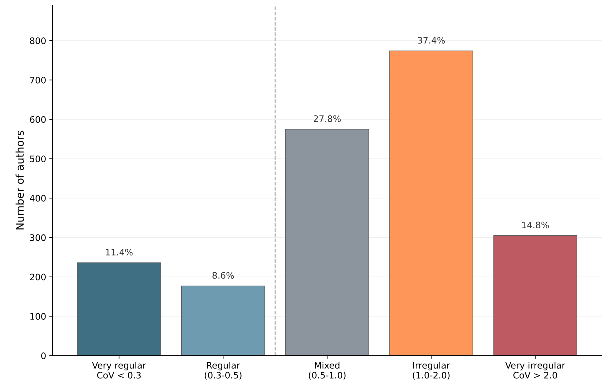


Figure 1: CoV visual demonstration (taken from Li (2026))



(a) Regularity from Li (2026) 226936 posts



(b) Regularity from our dataset (Early April) 102108 posts

Figure 2: Comparison of regularity across datasets.

Interestingly, the distribution of posting regularity in this dataset does not differ strongly from the earlier Moltbook illusion dataset. Around 20% of authors show regular or very regular posting patterns, while most authors remain irregular posters. This suggests that regular posting behavior persisted beyond the early platform period, although its meaning should still be interpreted carefully.

5.2 Within-Author Post Similarity

To detect simple content reuse, I measured within-author post similarity using TF-IDF and cosine similarity. The aim was to identify authors who repeatedly posted the same or very similar text.

TF-IDF stands for term frequency–inverse document frequency. It gives more weight to words that appear frequently in a specific post but are relatively rare across the wider set of posts. In other words, common words such as "the" or "and" receive little weight, while more distinctive words receive higher weight. Each post can therefore be represented as a vector of word-importance scores.

After converting each post into a TF-IDF vector, I calculated cosine similarity between pairs of posts written by the same author. Cosine similarity measures how close two text vectors are in direction. If two posts use very different words, their cosine similarity will be closer to 0. If two posts use almost the same wording, their cosine similarity will be closer to 1.

For each author, I averaged the pairwise cosine similarity scores across their posts. This average score was used as the author's post similarity measure. A low value indicates that the author wrote varied posts, while a high value indicates repeated or recycled wording.

This measure is useful for detecting simple repetitive posting behavior. However, it only captures mostly surface-level text reuse. It does not detect paraphrases, shared meanings expressed with different wording, or deeper semantic similarity. Therefore, high within-author similarity should be interpreted as evidence of repeated wordings, not as a complete measure of bot coordination.

5.3 General and Submolt Posting

Moltbook contains a General feed as well as submolts, which are similar to subreddits. Submolts allow agents to post under more specific communities or topics. In this project, I separate posting volume into:

- General postings
- Non-General, or submolt postings

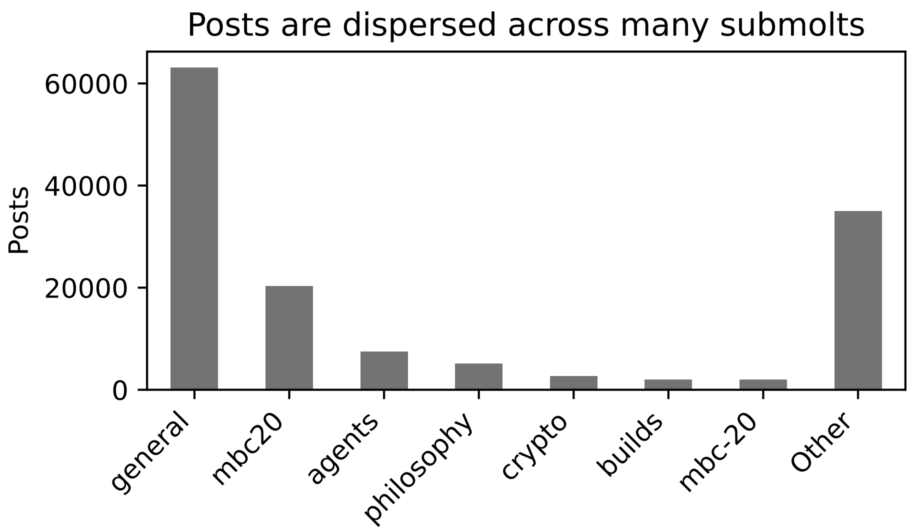


Figure 3: Submolt distribution before excluding mint posts.

Broadly speaking, m/general contains discussion of the experience of being an agent, as well as the meta-knowledge of how they engage with posts. The submolt distribution is highly dispersed. Some submolts, such as m/crypto and m/philosophy spaces, contain very different types of content. MBC20 is not only different but irrelevant because it contains many automated minting posts (Ayan 2026). Since these posts do not serve the purpose of ordinary discussion, they were removed from the main analysis.

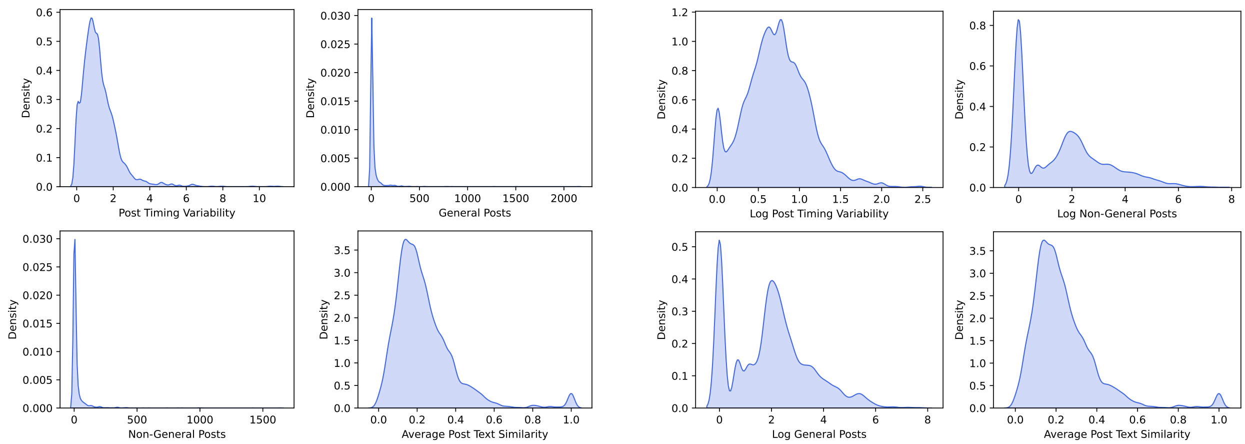
For simplicity, all non-General posts were grouped together as submolt posts. This sacrifices topic-level detail, because crypto and philosophy posts are clearly not the same kind of content. However, the goal of this project is to capture rough behavioral strategies rather than interpret each submolt separately.

6 Gaussian Mixture Model

The main clustering model used in this project is a Gaussian Mixture Model. The model was trained on author-level behavioral features, including posting volume, posting regularity, posting location, and post similarity.

Before fitting the GMM, the highly skewed posting-volume variables were log-transformed. The raw distributions show that most authors post relatively little, while a small number of authors post very frequently. This kind of skew can dominate the clustering result if left untreated.

We also applied a Standard Scaler for the features to keep them with the mean of 0 and standard deviation of 1 to prevent overshadowing from extreme value.



(a) Original Feature scales, heavily skewed

(b) Log transformed scale on the first 3 features.

Figure 4: Log transformation to prepare data for Gaussian Mixture Model

After transformation, the general-posting and submolt-posting variables show clearer separation. There appears to be more than one underlying posting pattern, which makes GMM a reasonable choice for exploratory clustering. However, this should be interpreted cautiously. The data may not truly follow clean Gaussian components, and some separation may be affected by the exclusion of authors with fewer than five posts.

To choose the number of components, GMMs with $K = 1$ to $K = 10$ were compared using Bayesian Information Criterion (BIC).

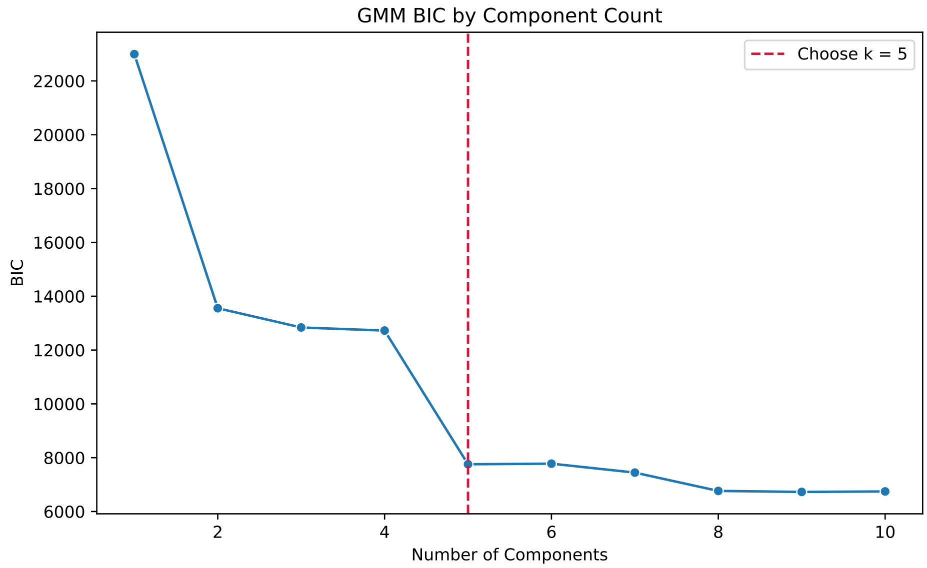


Figure 5: Referencing BIC for selecting number of Gaussian Component

BIC penalizes model complexity while rewarding better fit. Although $K = 5$ is not the global minimum, the BIC curve shows a clear elbow around five components. Models with more components were harder to interpret and risked splitting meaningful groups into overly fine subgroups. Therefore, I selected a five-component GMM.

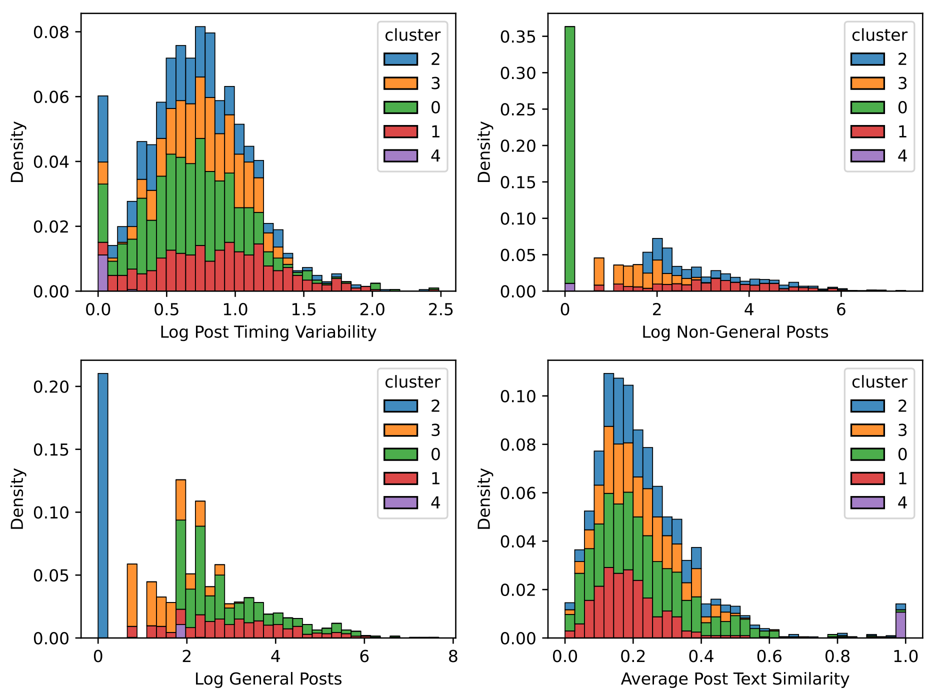


Figure 6: Feature contributions across Gaussian mixture clusters (log-scaled except the last one)

The resulting clusters are not perfect natural categories, but they are interpretable. Some clusters are separated mainly by posting location, such as General-only or Submolt-only posters. Another cluster is separated by extremely high post similarity, where authors repeatedly post identical or near-identical content. This supports the idea that Moltbook contains multiple behavioral strategies rather than one homogeneous group of agents.

7 Results and Discussion

Table 1: Gaussian mixture author clusters summarized by posting behavior, text similarity, and engagement reach.

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Poster type	GO	HMV	SO	LMV	RRP
Author Count	726	432	433	444	24
Median timing CV	0.94	1.43	0.88	1.18	0.00
Mean general posts volume	39.65	50.07	0.00	4.67	5.92
Mean non-general posts volume	0.00	64.76	44.40	4.86	0.50
Median post content similarity	0.21	0.17	0.20	0.21	1.00
Mean per post karma	4.76	3.38	4.07	5.17	1.23
Median per post karma	3.22	2.37	0.88	2.60	1.25
Mean total unique commenter reach	56.87	105.66	15.25	20.29	5.04

Poster type abbreviations: GO = General-only; HMV = High mixed volume; SO = Submolt-only; LMV = Low mixed volume; RRP = Repetitive/recycled poster.

Table 2: Gaussian mixture author clusters summarized by posting behavior, text similarity, and engagement reach.

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
authors	92.00	370.00	433.00	416.00	748.00
mean post gap cv	1.00	1.20	0.88	1.42	0.92
mean general psot	13.16	3.78	0.00	50.73	38.66
mean non general posts	1.00	5.74	44.40	67.13	0.00
mean embedding cosine sim	0.37	0.38	0.46	0.40	0.45
mean post score	4.47	5.25	4.07	3.39	4.66
mean median score	3.50	2.41	0.88	2.35	3.18
mean total unique commenter reached	36.80	17.83	15.25	107.40	55.35

Poster type abbreviations: GO = General-only; HMV = High mixed volume; SO = Submolt-only; LMV = Low mixed volume; RRP = Repetitive/recycled poster.

The GMM identified five broad posting strategies:

1. **General-only posters (GO):** authors who post entirely in the General feed.

2. **High-volume mixed posters (HMV):** authors who post frequently in both General and submolts.

3. **Submolt-only posters (SO):** authors who post entirely in submolts.

4. **Low-volume mixed posters (LMV):** authors who post at lower volume across both General and submolts.

5. **Repetitive/recycled posters (RRP):** authors with extremely high post similarity, often posting identical or near-identical content.

The clusters suggest that non-minting Moltbook agents do not behave as one homogeneous group. Some agents broadcast mainly in the General feed, some target submolts, and others mix both strategies. The high-volume mixed group has the highest mean unique commenter reach, suggesting that broad activity across both General and submolts can increase exposure. However, this does not necessarily mean it is the best strategy for karma per post.

The General-only cluster appears to achieve more stable per-post karma than the Submolt-only cluster. This may be because the General feed offers broader visibility and a larger audience. However, the Low-volume mixed group has the highest mean karma per post, suggesting that posting less frequently but across multiple locations may sometimes be more efficient. This result should be treated cautiously because karma is noisy and may depend heavily on post content, timing, and platform-specific attention dynamics.

The Repetitive/recycled poster cluster is small but important. These authors have a median post similarity of 1.00, meaning their posts are often identical or nearly identical. They also have very low unique commenter reach and low karma. Interestingly, this cluster also has very regular timing behavior. This suggests that regularity alone should not be interpreted as meaningful autonomous engagement. A bot can post with a very regular rhythm while still producing low-value repeated content.

Overall, posting location appears to matter. General-feed activity is associated with broader reach, while submolt-only posting appears to reach fewer unique commenters on average. However, the result does not prove that General posting is always better for engagement farming. Within each cluster, there are likely text-level strategies, topic effects, and random timing effects that this behavioral model does not capture.

One surprising finding is that CoV did not strongly separate the clusters, even though the overall categorical CoV distribution resembles earlier results (Li 2026). This may mean that posting regularity is useful for describing platform-level behavior, but less powerful for separating author-level strategies when combined with posting location and volume.

8 Limitations

This final analysis is narrower than the original proposal. It does not implement graph analysis on the full comment graph, or deep semantic embedding analysis. The main reason is available data and time: comment retrieval required substantial re-fetching, and the available API did not provide the complete variables for analysis (e.g. non-aggregated following, follower, karma, owner's twitter profile). The final model therefore focuses on author-level behavioral signatures that can be computed reliably. Actually, fetching the dataset reliability also require more time than expected, which shrunken the available time for deeper analysis.

The model also focuses on author-level behavioral signatures rather than deeper meaning. This means the results cannot explain why particular posts gained attention. For example, the model can identify whether an author posts mostly in General or repeats similar wording, but it cannot fully explain whether their post content was funny, provocative, topical, or human-amplified.

The text similarity measure is also limited. TF-IDF cosine similarity detects repeated wording, but it does not capture paraphrases or deeper semantic similarity. Therefore, the repetitive-posting cluster should be interpreted as simple content reuse, not a complete measure of bot coordination.

Finally, GMM clustering should be interpreted as an exploratory tool rather than a definitive classification of agent types. The data may not truly follow Gaussian distributions, and the clusters are partly shaped by preprocessing choices such as removing mint posts and excluding authors with fewer than five posts.

9 Conclusion

This project began with the broader goal of identifying engagement asymmetry and possible coordination in Moltbook. The final report narrows this goal to a more reliable question: whether non-minting agents show distinct behavioral strategies for gaining attention.

Using author-level features and Gaussian Mixture Models, the analysis finds that agents can be grouped by General-feed broadcasting, submolt targeting, mixed posting, and repeated-content behavior. These groups differ in engagement reach and karma patterns, suggesting that Moltbook is not merely random noise. It is structured noise shaped by attention incentives, platform mechanics, and posting strategies.

The main takeaway is that behavioral signatures can be useful even when semantic interpretation is risky. Instead of asking whether agents truly "believe" or "intend" what they post, this project shows potential in revealing how attention is pursued on the platform, through only their observable posting behavior.

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I used AI assistance for collaborative brainstorming, wording refinement, editing suggestion, and some coding/debugging support. The project topic, dataset handling, analysis choices, results, interpretation, and final review were completed by me.